FAZLE RABBI

Ph.D. Researcher in Software Engineering | Concordia University | Montréal, Canada

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Research interests: trustworthy AI agents, LLMs for software engineering, code robustness, verification, automated program repair, and code translation.

EDUCATION

Ph.D., Software Engineering — Concordia University, Canada (Sep 2022–present; expected 2027). GPA: 4.00/4.30.

M.Sc., Software Engineering — University of Dhaka, Bangladesh (2020). GPA: 3.73/4.00. Thesis: code-comment inconsistency detection.

B.Sc., Software Engineering — University of Dhaka, Bangladesh (2018). GPA: 3.59/4.00.

RESEARCH EXPERIENCE

Graduate Research Assistant

Concordia University | Sep 2022–present

- Conduct empirical research on the reliability of code-generating and code-executing LLM systems, including translation, automated program repair, and software agents.
- Design and evaluate reproducible benchmarks, multilingual test pipelines, execution-based evaluation methods, and agent workflows combining planning, testing, and verification.
- Collaborate on LLM security, social bias, model robustness, specification alignment, and multi-agent software engineering.

RESEARCH GROUPS & AFFILIATIONS

- O-RISA Lab, Concordia University — research member in AI-assisted software engineering, reliability, testing, and automated repair.
- NSERC CREATE SE4AI trainee; participated in a research visit to Polytechnique Montréal in 2024.
- DSSE Research Group, Institute of Information Technology, University of Dhaka — former member.

RESEARCH PROJECTS

- BabelCoder: multi-agent code translation using specification alignment, testing, and refinement.
- Code-executing agent reliability: study of failures in tool-use agents, framework interactions, and evidence-driven verification.
- Code generation / repair robustness: semantics-preserving changes to prompts and buggy code, robustness benchmarks, and comparative evaluation.
- Bias & security in LLM-generated code: evaluation and mitigation of fairness and vulnerability risks.

PEER-REVIEWED PUBLICATIONS & PREPRINTS

Specification-Driven Code Translation Powered by Large Language Models: How Far Are We?
Soumit Kanti Saha*, Fazle Rabbi*†, Tri Minh Triet Pham, Song Wang, Jinqiu Yang. Journal of Systems and Software · JSS, 2026. Paper ↗
* Equal contribution · † Corresponding author

A Multi-Language Perspective on the Robustness of LLM Code Generation
Fazle Rabbi, Zishuo Ding, Jinqiu Yang. Empirical Software Engineering · EMSE, 2026. Paper ↗

An Exploratory Study on Fine-Tuning Large Language Models for Secure Code Generation
Junjie Li, Fazle Rabbi, Cheng Cheng, Aseem Sangalay, Yuan Tian, Jinqiu Yang. Empirical Software Engineering · EMSE, 2026. Paper ↗

HEJ-Robust: A Robustness Benchmark for LLM-Based Automated Program Repair
Fazle Rabbi, Jinqiu Yang. ACM International Conference on AI-Powered Software · AIware, 2026. Paper ↗

Beyond Translation Accuracy: Addressing False Failures in LLM-Based Code Translation
Fazle Rabbi, Soumit Kanti Saha, Jinqiu Yang. ACM International Conference on AI-Powered Software · AIware, 2026. Paper ↗

Secure-Instruct: Prompt, Synthesize, and Fine-Tune for Secure Code Generation
Junjie Li, Fazle Rabbi, Bo Yang, Song Wang, Jinqiu Yang. International Conference on Predictive Models and Data Analytics in Software Engineering · PROMISE, 2026. Paper ↗

Bias Unveiled: Investigating Social Bias in LLM-Generated Code
Lin Ling, Fazle Rabbi, Song Wang, Jinqiu Yang. AAAI Conference on Artificial Intelligence · AAAI, 2025. Paper ↗

DVC in Open Source ML-Development: The Action and the Reaction
Lorena Barreto Simedo Pacheco, Musfiqur Rahman, Fazle Rabbi, Pouya Fathollahzadeh, Ahmad Abdellatif, Emad Shihab, Tse-Hsun (Peter) Chen, Jinqiu Yang, Ying Zou. International Conference on AI Engineering · CAIN, 2024. Paper ↗

SCMA: A Lightweight Tool to Analyze Swift Projects
Fazle Rabbi, Syeda Sumbul Hossain, Mir Mohammad Samsul Arefin. Software Engineering and Knowledge Engineering · SEKE, 2022. Paper ↗

SYSMAP: A Lightweight Software Visualization Tool to Analyze the Software Evolution of a System
Fazle Rabbi, Nishat Tasnim Niloy, Nadia Nahar, Md. Nurul Ahad Tawhid. arXiv preprint, 2021. Paper ↗

Detecting Code Comment Inconsistency Using Siamese Recurrent Network
Fazle Rabbi, Md. Saeed Siddik. International Conference on Program Comprehension · ICPC, 2020. Paper ↗

An Ensemble Approach to Detect Code Comment Inconsistencies Using Topic Modeling
Fazle Rabbi, Md. Nazmul Haque, Md. Eusha Kadir, Md. Saeed Siddik, Ahmedul Kabir. Software Engineering and Knowledge Engineering · SEKE, 2020. Paper ↗

MANUSCRIPTS UNDER REVIEW / REVISION

BabelCoder: Agentic Code Translation with Specification Alignment
Fazle Rabbi, Soumit Kanti Saha, Tri Minh Triet Pham, Song Wang, Jinqiu Yang. Major revision · ACM Transactions on Software Engineering and Methodology (TOSEM). Preprint ↗

Social Bias in LLM-Generated Code: Benchmark and Mitigation
Fazle Rabbi, Lin Ling, Song Wang, Jinqiu Yang. Under revision · Empirical Software Engineering (EMSE). Preprint ↗

TEACHING EXPERIENCE

Graduate Teaching Assistant — Concordia University (Jan 2024–present)

COMP 248 — Object-Oriented Programming

Tutorial instructor, lab instructor (LD), marker — Tutorials: Winter 2024; Winter, Summer, Fall 2026. Lab instructor and marker: Summer, Fall 2026.

SOEN 390 — Software Engineering Team Project

Stakeholder / project manager, tutorial instructor — Stakeholder and project manager: Winter 2024 and Winter 2025. Tutorials: Winter 2025.

SOEN 341 — Software Process and Practices

Tutorial, lab project evaluation, marker — Winter 2025.

COMP 352 — Data Structures and Algorithms

Marker — Fall 2024.

INDUSTRY EXPERIENCE

Software Infrastructure Engineer

Samsung R&D Institute Bangladesh | Feb 2020–Aug 2022

- Developed Jenkins-based CI/CD workflows for static analysis, quality metrics, and process compliance.
- Built internal software analysis and monitoring tools for multiple development teams.

Software Engineering Intern

Samsung R&D Institute Bangladesh | Jan–Jun 2018

- Implemented Android networking components and experimented with SMB and NetBIOS for file sharing.

ACADEMIC SERVICE & MENTORING

- Program committee: SANER 2027 Tool Demo Track.
- Journal reviewing: ACM Transactions on Software Engineering and Methodology (7 reviews); Empirical Software Engineering (2); Journal of Systems and Software (1).
- Research mentoring: two interns (2023 and 2026).
- Conference volunteering: SANER 2025.

TECHNICAL SKILLS

Programming: Python, C, C++, Java, Swift, JavaScript, TypeScript, PHP.

AI / data: PyTorch, Hugging Face Transformers, scikit-learn, LLM APIs, agent workflows, fine-tuning, test-based evaluation.

Systems: Git, Jenkins, CI/CD, static analysis, SonarQube, Linux, Bash.